

Pointwise convergence of multiple ergodic averages

Topic Summary · About 43 minutes

One starting point, different speeds.

When do long-term orbit averages
converge almost everywhere to a common product limit?

$$A_N^{\mathbf{b}}(x) = \frac{1}{N} \sum_{n=1}^N \prod_{j=1}^d f_j(T^{b_j n} x).$$

From oscillation control to joint distributions at selected lengths

The main claims

Original statements · Fixed functions and all integer lengths

Let $(X, \mathcal{F}, \mu)$ be a probability space, with T invertible, bimeasurable, probability-preserving, and mixing. Fix complex-valued $f_j \in L^\infty(\mu)$.

$$\mu(E \cap T^{-n}F) \longrightarrow \mu(E)\mu(F) \quad (|n| \rightarrow \infty).$$

Three positions

Fixed pairwise distinct nonzero integers
 b_1, b_2, b_3 , of either sign.

Any finite number of positions

Fixed $d \geq 2$,
 $(b_1, \dots, b_d) = (1, \dots, d)$.

$$A_N^b(x) \longrightarrow \prod_{j=1}^d \int_X f_j \, d\mu \quad \text{a.e.,} \quad N \in \mathbb{N}, \, N \rightarrow \infty.$$

All integer lengths; no mixing rate; no standardness assumption. Exceptional sets may depend on the fixed functions.

Direct consequence of A: any fixed finite set of distinct nonzero integer slopes of one sign, with all proof assumptions of A retained.

A shared starting point imposes a diagonal constraint

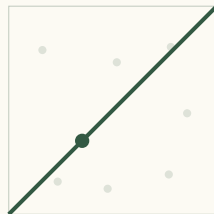
First difficulty: exceptional lengths along an orbit

$$S = T^{b_1} \times \cdots \times T^{b_d}, \quad F = f_1 \otimes \cdots \otimes f_d.$$

$$A_N^b(x) = \frac{1}{N} \sum_{n=1}^N F(S^n(x, \dots, x)).$$

The natural product-space measure is $\mu^{\otimes d}$.

The initial distribution here is $\Delta_* \mu$, where $\Delta(x) = (x, \dots, x)$.



In the nonatomic case, the diagonal may have product measure zero.

A Birkhoff full-measure set for $\mu^{\otimes d}$ need not control the diagonal initial distribution.

What does norm convergence leave uncontrolled?

The first difficulty

Overall error at a fixed length

$$\|A_N - L\|_2 \longrightarrow 0.$$

At each sufficiently late fixed length,
large errors occupy a small set.

The typewriter sequence: set $u_1 = 0$. For $k \geq 1$ and $0 \leq j < 2^k$, let

$$u_{2^k+j} = \mathbf{1}_{[j2^{-k}, (j+1)2^{-k})}, \quad \|u_{2^k+j}\|_2 = 2^{-k/2}.$$

Yet every $x \in [0, 1)$ satisfies $\limsup_n u_n(x) = 1$ and $\liminf_n u_n(x) = 0$.

The tail at a fixed point

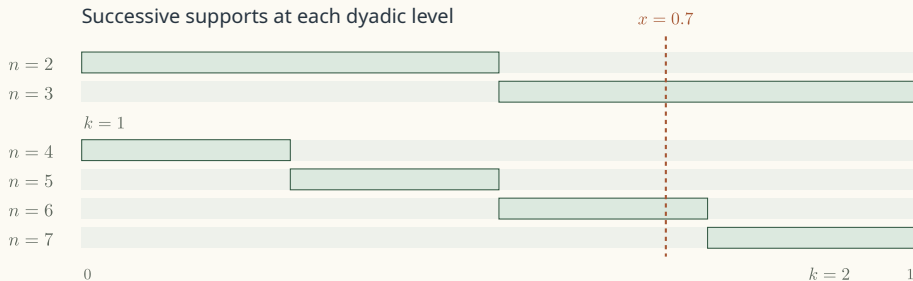
$$\sup_{M, N \geq K} |A_M(x) - A_N(x)| \longrightarrow 0.$$

At the same starting point,
bad lengths eventually stop recurring.

What does norm convergence leave uncontrolled?

The first difficulty

$$u_{2^k+j} = \mathbf{1}_{[j2^{-k}, (j+1)2^{-k})}, \quad \|u_{2^k+j}\|_2 = 2^{-k/2}.$$



Each row is one indicator. Every point is hit once at every level and missed on the other rows. This separates norm and pointwise convergence.

A convergent limit may still depend on the starting point

Existence and identification of the limit

Take the two-point probability space $X = \{0, 1\}$, with each point of probability $1/2$.

$$T(x) = x + 1 \pmod{2}, \quad f(x) = (-1)^x.$$

$$f(T^n x) f(T^{2n} x) f(T^{3n} x) = (-1)^{3x+6n} = f(x).$$

Stable at every length

The product of integrals is zero

$$A_N(x) = f(x).$$

$$\left(\int_X f \, d\mu \right)^3 = 0.$$

This system is invertible, measure-preserving, and ergodic, but not mixing. Structure can remain in the limit.

Earlier structural results show two possible behaviors

Background

Measure-distal systems

An invertible, ergodic Lebesgue probability system;
any fixed finite d and $f_j \in L^\infty$.

The averages along $n, 2n, \dots, dn$
converge almost everywhere.

Huang–Shao–Ye, Theorem C

Distal pointwise convergence will also determine parameter laws. PID is a separate additional assumption.

General norm convergence and these pointwise structural results supply different information.

Weak mixing and PID

An invertible Lebesgue probability-preserving
system, weakly mixing and PID; any fixed finite d
and bounded inputs.

The same averages converge almost everywhere
to the product of the integrals.

Gutman–Huang–Shao–Ye, Theorem 3.4

Two obstacles in the proof

The overall argument

1. Repeated excursions across scales

For every deterministic partition,
total oscillation grows sublinearly.

upper bound $C\sqrt{w}$ versus lower bound cw .

The triple results supply pointwise inputs; selected laws and structure organize the higher-order argument.

Norm limits enter both parts, identifying limits and removing fixed-length background contributions.

2. Bias from point-dependent lengths

Preserve failure in a joint distribution,
then prove enough conditional constraints
survive.

minimal support k , then maximize r .

Oscillation uniformly over deterministic partitions

From norm convergence to pointwise convergence

Any invertible probability-preserving system; fixed distinct nonzero integers b_j , with $\gcd(|b_1|, |b_2|, |b_3|) = 1$; $|f_j| \leq 1$. Fix an admissible smooth averaging kernel ϕ .

$$V_k^\phi(x) = \sum_{n \in \mathbb{Z}} 2^{-k} \phi(n/2^k) \prod_{j=1}^3 f_j(T^{b_j n} x), \quad k \geq 0.$$

For every deterministic choice $0 \leq k_1 < \dots < k_{w+1}$, the paper states

$$\sum_{i=1}^w \int_X \max_{k_i \leq \ell \leq k_{i+1}} |V_\ell^\phi - V_{k_i}^\phi| d\mu \leq C_{\phi, \mathbf{b}} \sqrt{w}.$$

The constant is independent of the system, inputs, endpoints, and number of blocks.

It may depend on the fixed kernel and slopes.

Neither mixing nor ergodicity is assumed here.

Sublinear oscillation rules out pointwise failure

A complete measure-theoretic deduction

1. If convergence fails, there are $\delta > 0$ and $\mu(E) = d_0 > 0$ such that every tail has a fixed positive oscillation on E .
2. Choose **deterministic endpoints** successively so that

$$\int_X \max_{k_i \leq \ell \leq k_{i+1}} |V_\ell - V_{k_i}| d\mu \geq \frac{\delta d_0}{2}.$$

3. For every block count w ,

$$\frac{\delta d_0}{2} w \leq C\sqrt{w},$$

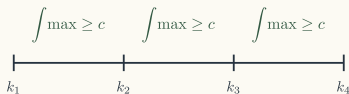
a contradiction as $w \rightarrow \infty$.

Sublinear oscillation rules out pointwise failure

A complete measure-theoretic deduction

$$cw \leq C\sqrt{w}, \quad c = \delta d_0/2 > 0.$$

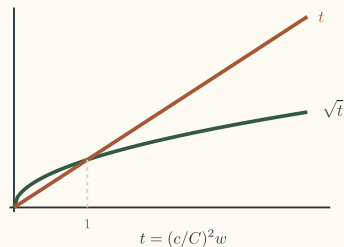
Successively chosen blocks



A positive cost at each chosen block.

One upper constant for every partition.

The two bounds conflict for large w



Divide both bounds by C^2/c . Their normalized values are t and $\sqrt{t}$. Fix the bad set and constants, then increase the number of chosen blocks.

Block maxima produce varying test functions

The local analysis behind oscillation

Linearizing the maximal oscillation makes the stopping scale depend on x :

$$\ell_i(x) \in \operatorname{argmax}_{k_i \leq \ell \leq k_{i+1}} |V_\ell(x) - V_{k_i}(x)|.$$

After correcting the phase, the difference kernel is paired with a depth-varying fourth input.

$$V_B(z) = \max_{s \in B} |z_s| + \sum_{s, s+1 \in B} |z_{s+1} - z_s| \leq W^{-1/2}.$$

The local quadrilinear estimate must be uniform in tree depth, block count, and the locations of input changes.

An operator bound for fixed functions does not yet supply this oscillation estimate.

Distinct slopes change the local geometry

The scope of the triple input

$$x, \quad x + b_1t, \quad x + b_2t, \quad x + b_3t.$$

The test position 0 and three observation positions form four linear forms.

Integer reconstruction

An integer left inverse uses the entire primitive configuration.

Quadratic cancellation

Vandermonde relations replace identities specific to arithmetic progressions.

Frequency counting

The height base respects divisibility by slope differences; dimension losses remain polynomial.

For $(1, 2, 4)$, replacing T by one common power and reordering does not reduce the slopes to $(1, 2, 3)$.

Mixing identifies a limit already known to exist

Combining norm and pointwise information

For fixed distinct nonzero integer slopes, mixing yields the norm limit

$$A_N^b \longrightarrow p = \prod_{j=1}^3 \int f_j \, d\mu \quad \text{in } L^2.$$

Hard truncation

Positive- and negative-time
partial sums

$$\|R_N^\pm\|_2 = o(N)$$

Oscillation gives existence; norm convergence identifies the product limit.

Smooth weights

Abel summation

$$V_k^\phi \rightarrow p \text{ in } L^2$$

The same limit

Take an a.e. convergent
subsequence
and compare with the existing
full-scale limit

All lengths require many ratios and one full-measure set

From smooth dyadic scales back to the original averages

For rational $c \in [1, 2]$, use admissible signed kernels to approximate

$$h_c(t) = c^{-1} \mathbf{1}_{(0, c]}(t), \quad \|\phi_{c, h} - h_c\|_1 < 2^{-h}.$$

Intersect the full-measure convergence sets of countably many kernels to obtain $A_{\lfloor c2^k \rfloor}(x) \rightarrow p$.

$$|A_M - A_N| \leq \frac{2|M - N|}{\max(M, N)} \quad (|F_n| \leq 1).$$

A finite rational grid in $[1, 2]$ controls neighboring lengths.

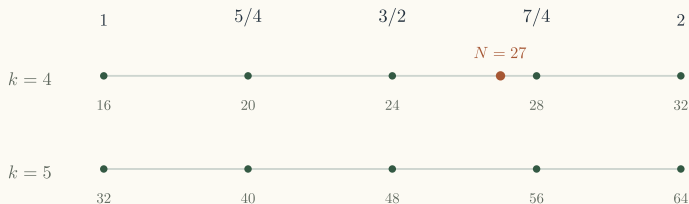
Fix the grid and kernel, then take the scale limit. Refine the grid and approximation only afterward.

All lengths require many ratios and one full-measure set

From smooth dyadic scales back to the original averages

$$|A_M - A_N| \leq \frac{2|M - N|}{\max(M, N)} \quad (|F_n| \leq 1).$$

The same rational ratios at every scale



Fix the grid and kernels. Take the scale limit. Then refine.

All rational ratios and kernel approximations use one countable intersection of full-measure sets. The displayed grid is one example. Fix it before the scale limit.

Higher-order failure yields a positive correlation

Second difficulty: distributions at selected lengths

After centering and selecting a real-valued term, failure yields a positive-measure set E and $\delta > 0$ such that

$$B_N(x) = \frac{1}{N} \sum_{n=1}^N \prod_{j \in J_0} f_j(T^{j^n} x) > 2\delta \quad \text{at arbitrarily late lengths.}$$

At each starting point, select a late length $N_a(x) \geq 2^a$ that preserves the deviation.

What is preserved

The limiting joint distribution of selected segments has a nonzero centered correlation.

The new difficulty

The length is chosen after seeing the starting point, so independence may change.

Higher-order failure yields a positive correlation

Second difficulty: distributions at selected lengths

$$F_1(a) = 1, \quad F_1(b) = -1, \quad F_2 = -F_1, \quad A_n = \frac{1}{n} \sum_{j=1}^n F_j, \quad n = 1, 2.$$

		$N = 1$	$N = 2$	Selected length
a	$\mu = 1/2$	1	0	$N(a) = 1$
b	$\mu = 1/2$	-1	0	$N(b) = 2$
Mean		0	0	$1/2$

A finite bounded-observation example of selection bias. The highlighted cells select $N(a) = 1$ and $N(b) = 2$. This is not a counterexample to an ergodic theorem.

Parameter laws and small marginals have different sources

The joint law of selected segments

On an extension Y of the unit-roof suspension, the selected law is

$$\int \prod_{i=1}^d G_i(y_i) d\lambda_Y = \lim_{a \rightarrow \mathcal{U}} \int_Y \frac{1}{N_a(\pi_Y y)} \int_0^{N_a(\pi_Y y)} \prod_{i=1}^d G_i(S_Y^{it} y) dt dm_Y.$$

Let W_Y be the maximal measure-distal factor of the time-one map, with $P_i = \omega_Y(y_i)$ and $P = (P_1, \dots, P_d)$.

The full parameter law

Arbitrary-order distal pointwise convergence makes the law of the entire P independent of the length selection.

Marginals of at most three states

$$\mathcal{L}((y_i)_{i \in J} \mid P) = \bigotimes_{i \in J} m_{Y, P_i}, \quad |J| \leq 3.$$

The required independence leaves one deliberate gap

Constraints before and after selection

Divide auxiliary outputs into three groups. Let x_0 be the original spatial coordinate, D the list of all output coordinates, D_H the sublist from groups H , and V all suspension phases.

$$\mathcal{L}(D \mid V) = \bigotimes_{\gamma} \mu.$$

$$\mathcal{L}(x_0, D_H \mid V) = \mu \otimes \bigotimes_{\gamma: \text{grp}(\gamma) \in H} \mu, \quad H \subsetneq \{1, 2, 3\}.$$

The original coordinate is independent of any two groups jointly.
It may still correlate with all three groups together.

Finite model: independent fair signs U_1, U_2, U_3 , with $X = U_1 U_2 U_3$.

Counting large tensor projections controls selection bias

Quantitative preservation of conditional laws

Suppose unit-ball vectors have at most L bad pairings per vector, and all other pairs satisfy $|\langle v_i, v_j \rangle| \leq \eta$. Then

$$\left\| \frac{1}{m} \sum_{j=1}^m v_j \right\|^2 \leq \frac{1+L}{m} + \eta.$$

The strengthening needed

Only a few $x_j \otimes y_j$
can have a large projection
onto the same low-rank test space.

Use in the selected law

Split into a high-covariance projection and a remainder.
Count the large projections and sum the remainder over candidate lengths.

Constants are independent of the ambient Hilbert-space dimension. No quantitative mixing rate is assumed.

Minimal support specifies the independence needed

From four positions to arbitrary finite order

At each position, the allowed variables are D , or (x_0, D_H) with $H \subsetneq \{1, 2, 3\}$.

First minimize support k

At fewer than k positions, the **allowed spatial lists** have their product law given all parameters.

Then maximize r

On this minimal support, maximize the number of positions using only auxiliary variables.

$$\mathbb{E}_{\lambda_Y} \left[\Phi(P) \prod_{i \in J} h_i(L_i(y_i)) \right] \neq 0, \quad |J| = k, \quad r < k.$$

Pointwise convergence for at most three terms on general systems still determines the small state marginals and gives $k \geq 4$. Minimality does not give product laws for arbitrary whole states.

From the other three states to a single state

Which conditional expectations are stabilized

Four positions

Condition on the parameters and the other three states.

$$\mathcal{D}_l = \sigma(P, (y_i)_{i \neq l}), \quad |I| = 4.$$

Both arguments also stabilize $\mathcal{D}_0 = \sigma(P)$.

Copying the fourfold “omit one row” argument would require the conditional law of the other $n - 1$ whole row states. Minimal support does not supply that law.

Approximate the supremum of bounded energies and take an inverse limit, so the specified conditional expectations persist under further extensions.

Arbitrary finite order

Condition on the parameters and one specified state.

$$\mathcal{D}_i = \sigma(P, y_i), \quad |I| = n.$$

Single-state stabilization connects rows and columns

One more position uses only auxiliary variables, on the same support

y_{11}	y_{1b}	$\cdots$
y_{b1}	y_{bb}	y_{bn}
$\cdots$	y_{nb}	y_{nn}

Row b : the new law's single-coordinate marginal yields the old law.

Column b : compatibility with the extension projection yields the old law.

The common corner: single-state stabilization gives the identity below.

No transpose symmetry is assumed.

$$\mathbb{E}[q((y_{ib})_i) \mid E, z_b] = g(p, y_{bb}).$$

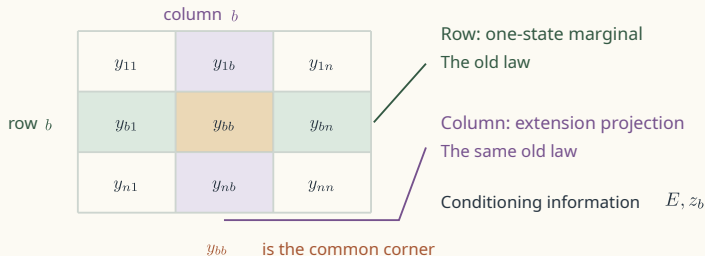
Together with the parameter laws and the nonvanishing argument, this preserves a nonzero correlation on k positions, while the number using only auxiliary variables increases from r to $r + 1$.

Single-state stabilization connects rows and columns

One more position uses only auxiliary variables, on the same support

$$\mathbb{E}[q((y_{ib})_i) \mid E, z_b] = g(p, y_{bb}).$$

Row and column share one corner



Only positions $1, b, n$ are shown. Intervening rows and columns are omitted. The diagram locates the conditioning information and does not assert transpose symmetry.

Two major dependencies determine the proof chain

The boundary of independent verification

Local quadrilinear analysis

H's local estimates and their depth-varying versions support triple oscillation.

Checks: uniform row and Bessel bounds, coefficient changes, and absorption of parameter losses.

Further checks concern nonergodic distal structure, small marginals conditional on all parameters, and compatibility of the saturated row–column identities.

The summary checks the main statements and these steps' roles. The full central proof chain has not been independently reviewed.

Rokhlin's higher-order mixing problem

Ordinary mixing is claimed to imply mixing of every finite order, used for higher-order correlations after length selection.

This is a major external result, not a formal consequence of the definition of mixing.

Two changes of quantifiers organize the problem

Discussion

From average error at each fixed length
to all late lengths at almost every starting point.

From correlations at deterministic lengths
to conditional laws that survive point-dependent selection.

Triple oscillation handles the first change.

Parameter laws, auxiliary independence, and the extremal array handle the second.

Further questions: which local uniformities are essential, and which conditional laws survive more general time configurations?

One-sign slopes: the direct consequence

Technical reference · Related to slide 2

Under the same assumptions on the system and functions as on main slide 2, A directly implies the same product limit for a finite set of distinct nonzero integer slopes of one sign. The verification has four steps:

1. If all $b_j > 0$, take $M = \max(2, \max_j b_j)$. At unused positions of $1, \dots, M$, insert the constant function 1 and apply A .
2. If all $b_j < 0$, set $S = T^{-1}$ and $c_j = -b_j > 0$.
3. The inverse is again invertible, bimeasurable, and probability-preserving, and $\mu(E \cap S^{-n}F) = \mu(T^{-n}E \cap F) \rightarrow \mu(E)\mu(F)$ as $|n| \rightarrow \infty$, so it is mixing.
4. Use $T^{b_j n} = S^{c_j n}$ and apply the first step.

This is a direct consequence of the stated theorem, retaining every assumption needed for its proof, rather than an additional theorem. The exceptional null set may depend on the fixed tuple and fixed slopes.

Repeated slopes and a standard factor

Technical reference · Related to slide 2

Distinctness matters. Repeated slopes require combining the corresponding functions first, so the limit is the product of the integrals of the combined functions. This consequence does not cover arbitrary higher-order mixed-sign slopes, arbitrary polynomials, or sparse times.

For the standard factor of a fixed tuple, choose bounded representatives and use the map $x \mapsto (f_j(T^n x))_{1 \leq j \leq d, n \in \mathbb{Z}}$. Its image distribution lies on a countable product of compact disks, and the shift intertwines with the transformation. The countable union of integer translates of a null set remains null. See A §2.1, p.3.

The admissible kernel and transference

Technical reference · Related to slide 8

Admissible averaging kernels

The kernel ϕ is real-valued, smooth, and compactly supported, with $\int \phi = 1$ and $\int t^r \phi(t) dt = 0$ for $r = 1, 2, 3$. There is $G \geq 1$ such that

$$\|\phi^{(r)}\|_{\infty} \leq [G(r+2)]^{G(r+2)} \quad (r \geq 0).$$

The kernel parameter G may depend on ϕ . The oscillation estimate is C Proposition 3.5, pp.8–9. If the original slopes have common divisor g , replace the transformation and slopes by T^g and b_j/g . In a general system, invertibility and measure preservation remain valid.

From the local estimate

The local quadrilinear real-line estimate passes through integer reconstruction and a finite-sequence estimate, then through truncated-orbit transference to an arbitrary probability system. Its constant is independent of the system, and no Rokhlin tower is required. See C §§3.1–3.2, pp.5–9, for the deduction.

Deterministic endpoints from nonconvergence

Technical reference · Related to slide 9

Quantifiers and endpoint selection

For a measurable complex sequence Z_k , define $\omega(x) = \inf_K \sup_{m,n \geq K} |Z_m(x) - Z_n(x)|$. If nonconvergence occurs on a positive-measure set, choose $E = \{\omega > 2\delta\}$ and $d_0 = \mu(E) > 0$. For each deterministic K , the triangle inequality yields $\sup_{\ell \geq K} |Z_\ell - Z_K| \geq \delta$ on E .

The finite maximum increases monotonically with L . By monotone convergence, some deterministic $L > K$ satisfies $\int \max_{K \leq \ell \leq L} |Z_\ell - Z_K| d\mu \geq \delta d_0/2$. Choose the endpoints recursively.

The essential input is an $o(w)$ upper bound uniform over every deterministic partition. A bound of only Cw admits the counterexample $Z_k = (-1)^k$. The contradiction also fails if the constant may depend on the endpoints.

The local estimate: kernel hypotheses

Technical reference · Related to slide 10

The full local hypotheses

Fix a primitive configuration of distinct integers $b_0 = 0, b_1, b_2, b_3$ and parameters $\delta_0 > 0, C_0 \geq 1$. On a finite dyadic tree rooted at J , take any subfamily $\mathcal{I}$. Each interval of length r , denoted I , has a kernel $w_I \in C_c^\infty(\mathbb{R}^2)$. On its support, each $x + b_j t$ lies in I at distance at least $\delta_0 r$ from the boundary. For every v, j , require $\int w_I(v - b_j t, t) dt = 0$, and impose the derivative bound $\|\partial_x^a \partial_t^d w_I\|_\infty \leq r^{-(a+d)} [C_0(a+d+2)]^{C_0(a+d+2)}$.

The local estimate: varying functions

Technical reference · Related to slide 10

Define $H_I(z_0, \dots, z_3) = r^{-2} \int_{\mathbb{R}^2} w_I(x, t) \prod_{j=0}^3 z_j(x + b_j t) \, dx \, dt$. Each input $z_{j,s}$ is specified at every depth and partitioned into W_j deterministic consecutive blocks, satisfying $V_B(z_j) \leq W_j^{-1/2}$ pointwise. Extend it by zero outside J . C Theorem 2.2 then asserts

$$\sum_{I \in \mathcal{I}} |I| |H_I(z_{0,s(I)}, \dots, z_{3,s(I)})| \leq C_{\mathbf{b}, \delta_0, C_0} |J|.$$

The constant is independent of grid translation, tree, depth, subfamily, kernels, and all four partitions and block counts. In the oscillation application, only the test slot among the four inputs varies. The central dependencies are D Theorem 3.2, C §1.2 and §§4–6, and the companion paper H.

Abel summation at both time directions

Technical reference · Related to slide 12

Abel summation at positive and negative times

Set $F_n = \prod_{j=1}^3 f_j \circ T^{b_j n}$ and $R_N^\pm = \sum_{n=1}^N (F_{\pm n} - p)$; then $\|R_N^\pm\|_2 = o(N)$. Fix $\phi \in C_c^1(\mathbb{R})$ supported in $[-B, B]$. Take $a \rightarrow \infty$, $M = \lfloor Ba \rfloor + 1$, and $w_n^\pm = a^{-1} \phi(\pm n/a)$, so that $w_M^\pm = 0$.

$$\sum_{n=1}^M w_n^\pm (F_{\pm n} - p) = \sum_{n=1}^{M-1} (w_n^\pm - w_{n+1}^\pm) R_n^\pm.$$

The norm of the right side is at most $a^{-2} \|\phi'\|_\infty \sum_{n < M} \|R_n^\pm\|_2$. First choose N_0 so that the tail satisfies $\|R_n^\pm\|_2 \leq \varepsilon n$. The finite initial part multiplied by a^{-2} tends to zero; the tail has limit superior $O_{\phi, B}(\varepsilon)$. Finally let $\varepsilon \rightarrow 0$. The $n = 0$ term is $O(a^{-1})$, and the constant term tends by Riemann summation to p . Taking $a = 2^k$ proves the conclusion.

The norm input includes slopes of mixed signs; see C Lemmas 3.7–3.8, pp.10–11. Smooth weighted convergence should not be described as an immediate consequence of L^2 convergence for the hard averages without this argument.

One full-measure set and all lengths

Technical reference · Related to slide 13

Set $N_k = \lfloor c2^k \rfloor$ and normalize by $|F_n| \leq 1$. For a fixed admissible kernel ϕ ,

$$|A_{N_k} - V_k^\phi| \leq \frac{1}{c2^k} + 2^{-k} \sum_{n \in \mathbb{Z}} |h_c(n/2^k) - \phi(n/2^k)|.$$

The second term is a Riemann sum for a compactly supported function with finitely many discontinuities, and tends to $\|h_c - \phi\|_1$. Thus, on the common full-measure set, $\limsup_k |A_{N_k} - p| \leq 2^{-h}$; then let $h \rightarrow \infty$.

Choose a finite rational grid containing 1, 2, with mesh at most η . For $2^k \leq N < 2^{k+1}$, arrange $|N - \lfloor c2^k \rfloor| \leq \eta 2^k + 1$. The length-comparison error is at most $2\eta + 2^{1-k}$. First take $k \rightarrow \infty$, then $\eta \rightarrow 0$.

A Lemma 2.5 combines smooth oscillation with the norm limit on general systems to obtain full-length almost-everywhere convergence for at most three distinct positive integer slopes on a standard invertible probability-preserving system. Mixing is not required there, and the limit is not asserted to be the product of the integrals.

Selected lengths and the suspension

Technical reference · Related to slide 14

See A Lemma 3.1, pp.6–7. The choice $N_a(x, u)$ is independent of the suspension phase u and takes values in $\{c2^k : c \in F, k \geq a\}$, where $F \subset [1, 2]$ is a fixed finite grid. On E^c , it equals 2^a .

Passing the discrete positive correlation to the suspension flow also requires a nonnegative continuous phase test Ψ . It restricts the fractional time so that integer parts agree across slots, allowing comparison of flow and discrete averages with error $O(2^{-a})$. The background outside the set vanishes by deterministic-length L^2 convergence to zero. Thus the retained positive correlation is not obtained by an unconditional substitution of continuous time for discrete time.

At this stage, the arbitrary-order argument needs only some nonempty centered support J_0 ; it does not assume that all d slots occur. Minimal support is selected later.

The selected law on an extension

Technical reference · Related to slide 15

The suspension and the selected law

$C = X \times [0, 1)$, $m_C = \mu \otimes du$, and $S_C^t(x, u) = (T^{\lfloor u+t \rfloor} x, \{u+t\})$. The time-one map preserves the phase u and is therefore generally nonergodic. The flow Y is jointly Borel and probability-preserving on a standard probability space, with a probability-preserving factor map π_Y onto C . Compatibility at each fixed time is understood modulo null sets; joint measurability and Fubini justify time integration.

Fix the same family of selectors and a free ultrafilter $\mathcal{U}$. A Proposition 3.2, pp.7–8, establishes a unique law λ_Y satisfying the displayed identity for all bounded measurable slot tests. Its single-coordinate marginals are m_Y , it is invariant under $(y_i) \mapsto (S_Y^{it} y_i)$, and it is compatible with extensions and countable inverse limits.

Conditioning on the full parameter array

Technical reference · Related to slide 15

Conditioning on all parameters

Here W_Y denotes the maximal measure-distal factor of the time-one map. A Proposition 2.3 ensures that every flow time preserves it. After a fixed time shift of G_i , conditional expectations can therefore still be treated on the same factor.

The conditional law in A Proposition 3.3 is not deduced merely from a few slot marginals. One first obtains a relative product conditional on $(P_i)_{i \in J}$, then views that relatively weakly mixing system and the full distal parameter array as a joining over their common factor. Relative disjointness upgrades the conditioning to P . The nonergodic versions, stability under powers, and relative disjointness are the structural inputs in B Proposition 2.2 and A Proposition 2.3.

Auxiliary independence and the sign model

Technical reference · Related to slide 16

On the slide, $\text{grp}(\gamma)$ is the group number of output γ . The tensor product runs over all output coordinates whose group lies in H . A group may contain several outputs or be empty.

Each output $D_\gamma : Y \rightarrow C$ is probability-preserving and satisfies $D_\gamma(S_Y^t y) = S_C^{q_\gamma t} D_\gamma(y)$, with $q_\gamma > 0$ rational. Definition 4.1 and Lemma 4.2 of A, pp.10–11, also upgrade these conditions from the phase V to the maximal distal factor W_Y . This uses relative disjointness between the fixed-phase mixing product system and a distal extension.

For the finite example, conditional on any two signs, the third remains independent and fair, so the conditional law of X is still fair. Yet $XU_1U_2U_3 = 1$ holds identically, ruling out joint independence of all four variables.

The dimension-independent tensor estimate

Technical reference · Related to slide 17

The tensor estimate

B Theorem 4.1, p.11, and A Theorem A.1, p.21: given $0 < \varepsilon \leq 1$ and integers $d \geq 1, L \geq 0$, let $x_1, \dots, x_m$ and $y_1, \dots, y_m$ lie in the unit balls of two real Hilbert spaces. For each family, suppose a graph of maximum degree at most L contains every off-diagonal pair whose absolute inner product exceeds η , with $0 \leq \eta \leq \varepsilon^2/10^4$. The two graphs may differ. Every orthogonal projection of rank at most d , denoted P , then satisfies

$$\#\{j : \|P(x_j \otimes y_j)\| \geq \varepsilon\} \leq \frac{C(1+L)}{\varepsilon^2} \exp\left(C\varepsilon^{-2}\sqrt{d} \log(2+d)\right).$$

The constant C is universal. For $d_m = \lceil (1 + \log m)^{3/2} \rceil$, the bound is $o_{\varepsilon,L}(m)$, uniformly in the ambient dimensions, the projection, and admissible η . The proof also uses separation of high-covariance directions and Gaussian comparison. The square expansion on main slide 17 is only a basic observation within it.

Summing the selected-length remainder

Technical reference · Related to slide 17

The selected-length argument is A Proposition 4.3, pp.11–14. Fix an integer h clearing the speed denominators, let $m(k, c) = \lfloor c2^k/h \rfloor$, and let $p_{k,c}$ be the conditional probabilities of the selection events. Then $\sum_{k,c} p_{k,c} = 1$. For sufficiently large a , the remainder satisfies

$$\sum_{\substack{k \geq a \\ c \in F}} \sqrt{\frac{p_{k,c}}{d_{m(k,c)}}} \leq \left(\sum_{\substack{k \geq a \\ c \in F}} d_{m(k,c)}^{-1} \right)^{1/2} \longrightarrow 0.$$

Here $d_{m(k,c)}^{-1} = O(k^{-3/2})$; the series inside the square root is summable. For the projection part, first fix ε and take the selection-scale limit, then let $\varepsilon \rightarrow 0$.

Minimal support: allowed spatial lists

Technical reference · Related to slide 18

The exact choices are in A §5.1 and Lemma 5.1, pp.14–15. The function Φ is a bounded real function of the full distal parameter array. Each h_i is a bounded real test of its assigned spatial list, centered for the corresponding spatial product law. The allowed lists are $L_i = D$, or $L_i = (x_0, D_{H_i})$, where $H_i \subsetneq \{1, 2, 3\}$. Conditional independence also centers these tests over their own distal parameters.

Assign one allowed list to each of fewer than k distinct positions. Expand each test into its mean and centered part. Every nonconstant term has support smaller than k and vanishes by minimality, giving the spatial product law conditional on all parameters. This does not permit replacing the lists by arbitrary measurable whole states. Separately, A Proposition 3.3 uses pointwise convergence for at most three terms on general systems to determine the conditional joint laws of at most three whole states.

Three groups and the extremal choice

Technical reference · Related to slide 18

Choose a position b involving the original variable. Since $k \geq 4$, a surjection $\tau : J \setminus \{b\} \rightarrow \{1, 2, 3\}$ distributes each of the other positions' complete allowed lists among three groups. Omitting a new group omits at least one old support position, so the product law below k applies. The old group numbers at the corner are retained, keeping the original-variable test within the allowed class. The full new auxiliary product law also uses maximality of r ; see A Lemma 5.3, pp.16–17.

The fourfold proof maximizes r directly among four-position correlations. B Proposition 6.1 also uses an array and Bessel's inequality to exclude $r = 3$. The arbitrary-order proof does not copy that extra bound: minimal support and the full auxiliary product law need only give $r < k$.

The exact stabilization identity

Technical reference · Related to slide 19

Two choices of conditioning information

B Proposition 6.2 and (6.2)–(6.3), pp.19–20, use $\mathcal{D}_0(Y) = \sigma(P)$ and $\mathcal{D}_l(Y) = \sigma(P, (y_i)_{i \neq l})$, with exactly four positions. A Lemma 5.2 and (5.2)–(5.3), pp.15–16, instead use $\mathcal{D}_0(Y) = \sigma(P)$ and $\mathcal{D}_i(Y) = \sigma(P, y_i)$.

In each case, every standard flow over C has a standard flow extension Y_* such that, for every further standard flow extension $\rho : Y' \rightarrow Y_*$, every $f \in L^2(\lambda_{Y_*})$, and each $\ell \in \{0\} \cup I$, the corresponding paper's definition of $\mathcal{D}_\ell$ gives

$$\mathbb{E}_{\lambda_{Y'}} [f \circ \rho^I \mid \mathcal{D}_\ell(Y')] = \mathbb{E}_{\lambda_{Y_*}} [f \mid \mathcal{D}_\ell(Y_*)] \circ \rho^I.$$

Bounded energies and the inverse limit

Technical reference · Related to slide 19

The selected law is compatible with extensions and countable inverse limits, and stage tests are dense in the limit L^2 space. Extensions increase the conditioning information, so $\|\mathbb{E}[f \mid \mathcal{D}_\ell]\|_2^2$ is nondecreasing and bounded by $\|f\|_2^2$. For projections P_1, P_2 onto nested closed subspaces, $\|P_2 f\|^2 = \|P_1 f\|^2 + \|(P_2 - P_1)f\|^2$. Equal energies therefore force equal conditional expectations. The fourfold use is B Lemma 6.4, (6.9)–(6.11), pp.21–22. The single-state use at arbitrary order is A Lemma 6.1, (6.5), p.18. This change avoids first requiring pointwise convergence of $n - 1$ -term averages on general auxiliary systems. The pointwise result for at most three terms is still used.

The row–column law and its common corner

Technical reference · Related to slide 20

The precise row–column identity

Let $Z = M_b(Y)$, with space Y^I , measure $m_Z = \lambda_Y$, and flow $S_Z^t((y_i)_i) = (S_Y^{it/b} y_i)_i$. Projection onto slot b is therefore compatible with S_Y^t and is a flow factor. Next take $z_i = (y_{ij})_j$ and $E_i = \omega_Z(z_i)$. Write the column states as $y_i = y_{ib}$, their old parameter array as p , and the corner's old parameter as w . Set $q((y_i)_i) = \prod_{j \in J \setminus \{b\}} h_j(L_j(y_j))$, $g(P, y_b) = \mathbb{E}_{\lambda_Y}[q \mid P, y_b]$, and $u(e, y_b) = \mathbb{E}_{m_Z}[q \mid e, y_b]$. The original nonzero correlation implies $g \neq 0$ in L^2 .

A Lemma 6.1, pp.18–19, gives

$$\mathbb{E}[q((y_i)_i)q(z_b) \mid E] = \int_Y g(p, y)u(E_b, y) \, dm_{Y,w}(y).$$

Why the row–column pairing is nonzero

Technical reference · Related to slide 20

The stronger identity needed is $\mathbb{E}[q((y_i)_i) \mid E, z_b] = g(p, y_{bb})$, supplied by one-slot saturation. Lemma 6.2 treats the additional conditional independence of parameters. Conditional on w , the parameters p and $e = E_b$ are independent. Let $R(e)$ denote the old full parameter array determined by the new distal parameter. Their conditional laws α_w, β_w satisfy $R_*\beta_w = \alpha_w$ and $\mathbb{E}_{\beta_w}[u(e, \cdot) \mid R(e)] = g(R(e), \cdot)$. If the row–column pairing vanished identically, conditionally averaging e over $R(e)$ would make two independent identically distributed g vectors orthogonal almost everywhere. In a separable real Hilbert space this forces $g = 0$, contradicting the original correlation. A nonzero pairing therefore exists, producing a new correlation with one more position using only auxiliary variables. The quantity $\int gu$ has not simply been treated as a squared norm. See A §§6.2–6.4, pp.19–20. The array on main slide 20 only locates these conditional expectations.

The distinct fourfold argument

Technical reference · Related to slide 20

For comparison, B Lemma 6.4, (6.9)–(6.11), first omits one row, uses conditional independence of the other three whole row states, and then uses stabilization over the parameters and the other three states to restore the omitted column state. At arbitrary order, (6.5) gives the displayed single-state identity directly. It neither upgrades the minimal-support product laws for lists to independence of whole states nor removes the pointwise assumption for at most three terms on general systems.

The two external proof dependencies

Technical reference · Related to slide 21

The local analytic dependency

H: An L3 bound for the trilinear Hilbert transform (October 5, 2026). D Theorem 3.2 develops the depth-varying version; C Theorem 2.2 and §§4–6 treat distinct slopes. B has its own coefficient appendix, so the four papers do not form a simple linear sequence of proofs.

The exact higher-order mixing input

R: Rokhlin's multiple-mixing problem for one transformation (September 23, 2026). A Theorem 2.1 imports the following assertion: if T is an invertible mixing probability-preserving transformation, then, for each fixed $s \geq 2$, functions $g_j \in L^\infty$, and integer times,

$$\int_X \prod_{j=1}^s g_j(T^{t_j} x) \, d\mu \longrightarrow \prod_{j=1}^s \int_X g_j \, d\mu \quad \text{as } \min_{i \neq j} |t_i - t_j| \rightarrow \infty.$$

The independent proof review of this external manuscript is not complete here. Its use in A Theorem 2.1 and in B's selected-length argument must remain an explicit dependency. It cannot be replaced by unpacking the definition of ordinary mixing.

Sources and fixed versions

Technical reference · Related to slide 22

The four papers and the fixed version

The links below are official public sources. The summary uses the repository version checked on October 8, 2026, at commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Opening the original papers requires internet access; the offline summary and its formulas do not.

- Pointwise Multiple Ergodic Averages for Mixing Transformations (October 4, 2026, 25 pages).
- Pointwise convergence of fourfold ergodic averages for mixing transformations (October 4, 2026, 36 pages).
- Triple ergodic averages with distinct integer slopes (October 4, 2026, 43 pages).
- Pointwise convergence of triple ergodic averages for mixing transformations (October 4, 2026, 61 pages).

Published background: Huang–Shao–Ye, Theorem C; Gutman–Huang–Shao–Ye, Theorem 3.4.

The official repository and its verification-status notes. Public release is not treated here as equivalent to full peer review or formal verification.